

10.2 Access Control

Authentication with a Local Password

Many types of authentication can be performed on networking devices, and each method offers varying levels of security.

User accounts configured locally on each device is not scalable.

The method provides no fallback authentication method.

SSH is a more secure form of remote access:

- It requires a username and a password.
- The username and password can be authenticated locally.

```
R1(config)# line vty 0 4
R1(config-line)# password ci5c0
R1(config-line)# login
```

```
R1(config)# ip domain-name example.com
R1(config)# crypto key generate rsa general-keys modulus 2048
R1(config)# username Admin secret Str0ng3rPa55w0rd
R1(config)# ssh version 2
R1(config)# line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
```

Access Control

AAA Components



Authentication

Who are you?

Authorization

How much can you spend?

Accounting

What did you spend it on?

Account Number: 1234-567-890 Statement Closing Date: 01-31-01 Current Amount Due: \$278.50

JOE EMPLOYEE
456 SKYVIEW DRIVE
HOMETOWN, USA 99900-1234

MAIL PAYMENT TO:
THE BANK
132 YNL STREET
ANYTOWN, USA 67500-0010

872919345 00178255000000003

Detach here and return upper portion with check or money order. Do not staple or fold.

Statement of Personal Credit Card Account
Retain this portion for your files.

Cardmember Name: JOE EMPLOYEE Account Number: 1234-456-890 Statement Closing Date: 01-31-01

Statement Date: 02-01-01 Payment Due Date: 03-01-01

Closing Date: 01-31-01

Credit Limit: \$1,500.00 Credit Available: \$1221.50

New Balance: \$278.50 Minimum Payment Due: \$20.00

Account Summary

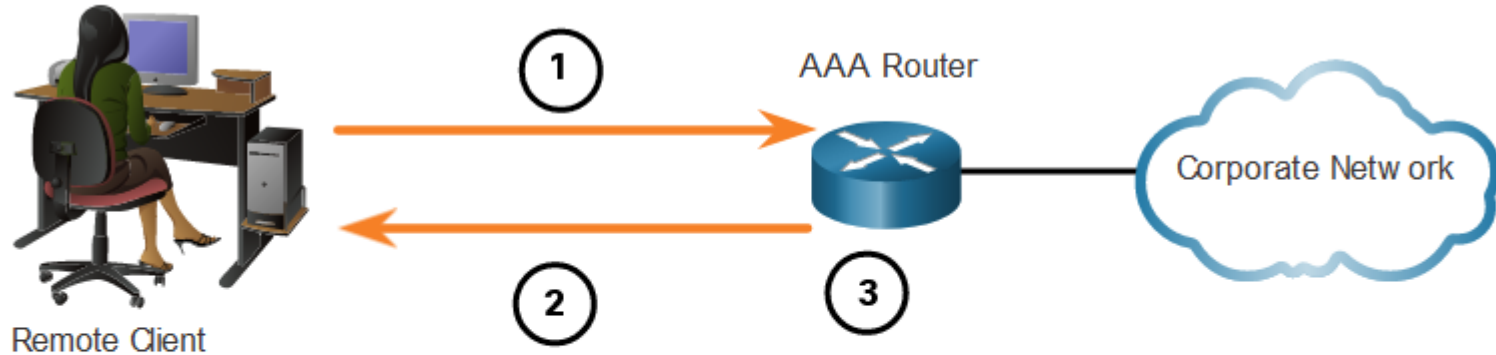
Previous Balance:	+74.24	Transaction Fees:	+3.00
Purchases:	+250.50	Annual Fees:	+25.00
Cash Advances:	+0	Current Amount Due:	+250.50
Payments:	-74.25	Amount Past Due:	+0
Finance Charge:	+0	Amount Over Credit Line:	+0
Late Charge:	+0	NEW BALANCE:	\$278.50

Reference Number	Sold	Posted	Activity Since Last Statement	Amount
43210987	01-03	01-13	Payment, Thank You	-\$74.25
01234567	01-12	01-13	Wings 'N' Things Anytown, USA	\$25.25
78901234	01-14	01-17	Record Release Anytown, USA	\$40.00
45678901	01-14	01-17	Sports Stadium Anytown, USA	\$75.25
3210987	01-22	01-23	Tie Tack Anytown, USA	\$20.75
76543210	01-29	01-30	Electronic World Anytown, USA	\$89.25
2345678		01-30	Transaction Fees	\$3.00
34567890		01-01	Annual Fee	\$25.00

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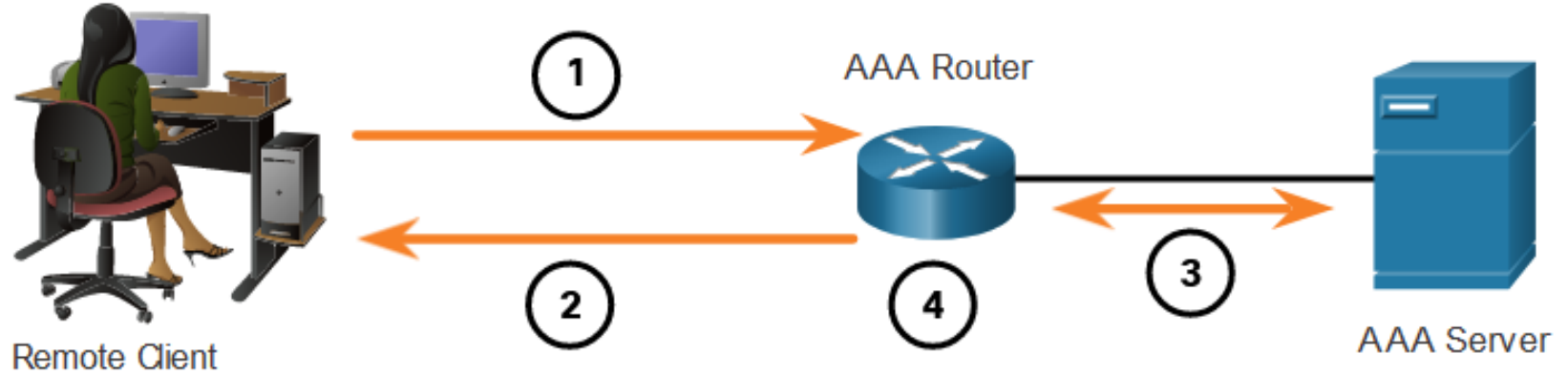
Access Control Authentication

Local and server-based are two common methods of implementing AAA authentication.



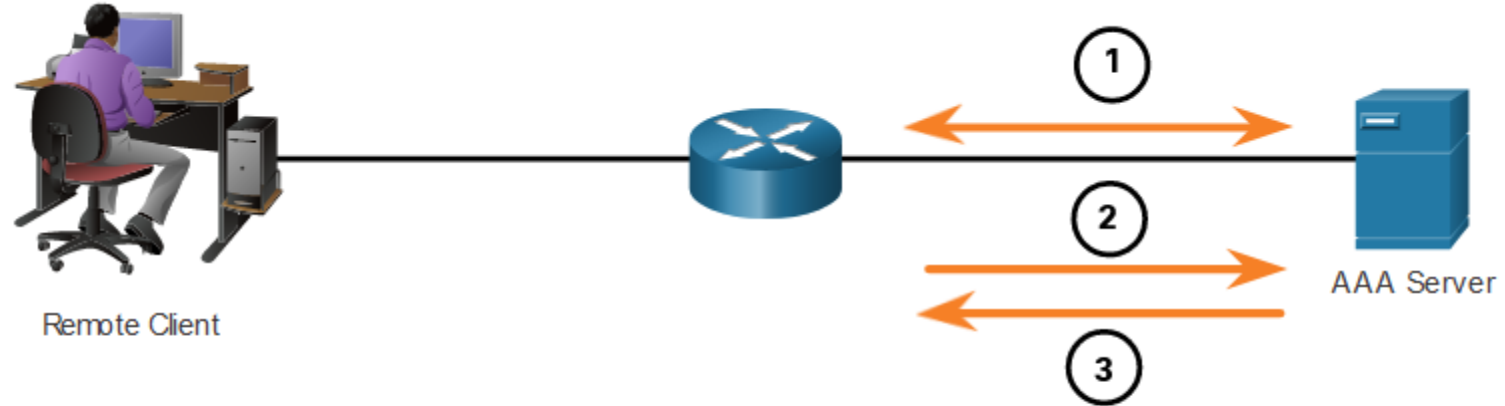
1. The client establishes a connection with the router.
2. The AAA router prompts the user for a username and password.
3. The router authenticates the username and password using the local database and the user is provided access to the network based on information in the local database.

Access Control Authentication



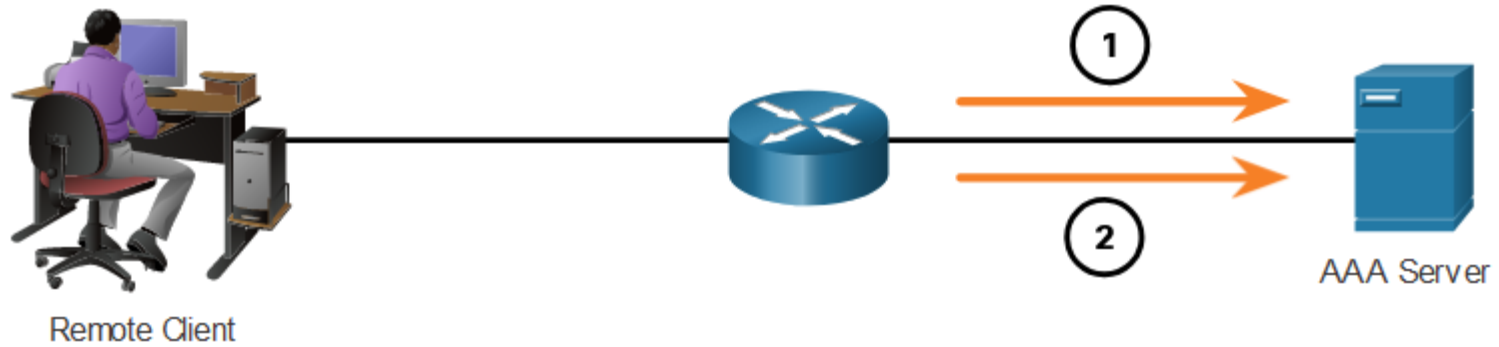
1. The client establishes a connection with the router.
2. The AAA router prompts the user for a username and password.
3. The router authenticates the username and password using a AAA server.
4. The user is provided access to the network based on information in the remote AAA server.

Access Control Authorization



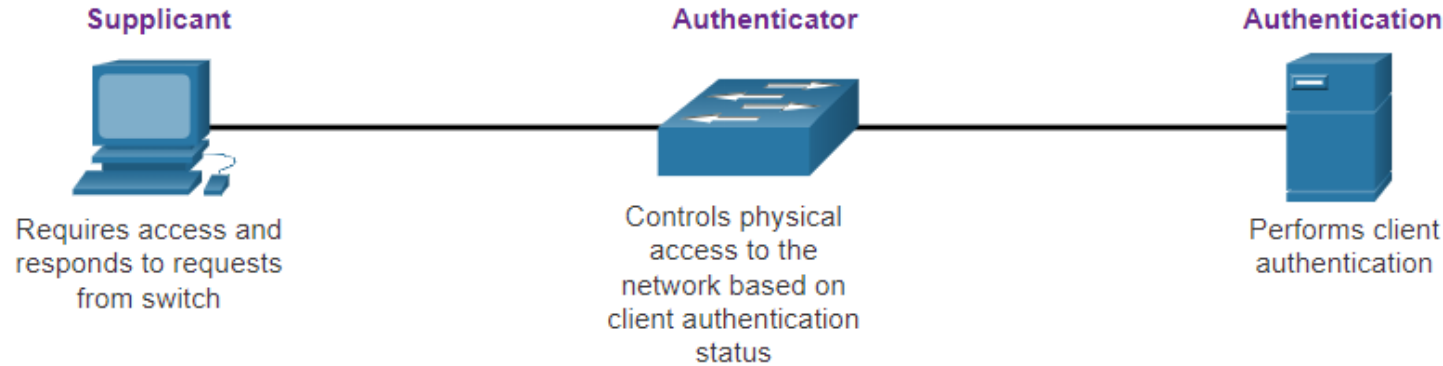
1. When a user has been authenticated, a session is established between the router and the AAA server.
2. The router requests authorization from the AAA server for the client's requested service.
3. The AAA server returns a PASS/FAIL response for authorization.

Access Control Accounting



1. When a user has been authenticated, the AAA accounting process generates a start message to begin the accounting process.
2. When the user finishes, a stop message is recorded and the accounting process ends.

Access Control 802.1X



- **Client (Supplicant)** - This is a device running 802.1X-compliant client software, which is available for wired or wireless devices.
- **Switch (Authenticator)** –The switch acts as an intermediary between the client and the authentication server. It requests identifying information from the client, verifies that information with the authentication server, and relays a response to the client. Another device that could act as authenticator is a wireless access point.
- **Authentication server** –The server validates the identity of the client and notifies the switch or wireless access point that the client is or is not authorized to access the LAN and switch services.